A

PBL REPORT

ON

"ROBUST POSTURE RECOGNITION USING ALIGNORA IN BANKING AND SOFTWARE SECTOR"

FOR PARTIAL FULFILLMENT
OF THE REQUIREMENTS FOR THE
PROJECT BASED LEARNING
OF S.E. E&CE – 2019 COURSE, SPPU, PUNE

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CERTIFICATE

This is to certify that the Project Based Learning Report entitled
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Is a bonafide work carried out by them under the supervision of Robust Posture Recognition Using Alignora In Banking And Software Sector. And it is approved for the partial fulfillment of the requirements for the Project based learning of S.E. E&CE – 2019 Course of the Savitribai Phule Pune University, Pune.

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ABSTRACT

This project research focused to presents a novel wearable posture detection system designed for comparative analysis of working professionals, aiming to address posture-related musculoskeletal issues. The system integrates an MPU6050 inertial measurement unit and an ESP32 microcontroller to capture real-time accelerometer and gyroscope data, enabling precise posture monitoring. A sensor fusion technique is implemented to enhance data accuracy by integrating multiple sensor inputs, improving posture detection precision .A unique algorithm processes the sensor data, detects deviations from optimal posture, and transmits the analyzed information via Bluetooth to an Android application for real-time feedback. Additionally, the system incorporates Firebase Realtime Database for cloud-based posture tracking, facilitating long-term analysis and comparative studies among users. Unlike conventional posture correction solutions, this system offers a lightweight, cost-effective, and continuously adaptive approach tailored for workplace environments. The proposed solution demonstrates high accuracy in detecting postural deviations, making it a valuable tool for ergonomics assessment, occupational health management, and injury prevention.

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CHAPTER 1

Introduction

1.1 Background

Posture-related health issues have been a growing concern in modern society, especially with the rise of sedentary lifestyles, long working hours, and increased screen usage. Poor posture can lead to musculoskeletal disorders (MSDs) such as back pain, neck strain, and spinal misalignment, significantly impacting an individual's health and productivity.

The importance of posture correction has been recognized in medicine, ergonomics, and workplace safety. Traditional methods for posture correction, such as physical therapy, ergonomic chairs, and manual posture reminders, have been in use for decades. However, these approaches often rely on human intervention and lack real-time feedback, making them less effective in maintaining proper posture throughout daily activities.

Historical Background:

- The concept of postural awareness dates back to ancient medical practices where body alignment was emphasized in yoga, martial arts, and traditional physical therapies.
- In the 20th century, the impact of poor posture gained attention in occupational health, leading to ergonomic furniture designs and posture-training programs.
- With advancements in wearable technology and IoT (Internet of Things), the development of smart posture correction devices began in the early 2010s. These devices used sensors like accelerometers, gyroscopes, and machine learning algorithms to detect and correct posture deviations.

1.2 Relevance

The Wearable Posture Detection System is highly relevant to the field of Electronics and Communication Engineering (ECE) as it integrates sensor technology, embedded systems, wireless communication, and cloud computing, all of which are core areas of the ECE curriculum.

- **Embedded Systems & Microcontrollers:** The system uses an ESP32 microcontroller, which is widely studied in ECE for real-time data processing and IoT applications.
- **Sensor Technology:** The MPU6050 IMU sensor (accelerometer and gyroscope) is used for motion detection, a fundamental concept in electronics and automation.
- **Wireless Communication:** The project employs Bluetooth technology for data transmission to an Android application, demonstrating real-world applications of wireless connectivity.
- **Cloud Computing & IoT:** Integration with Firebase Realtime Database enables cloud storage and remote monitoring, aligning with modern trends in IoT-based health monitoring systems.

1.3 Literature Survey

1. Wearable posture monitoring systems have gained popularity due to their potential for preventing musculoskeletal disorders (MSDs) and promoting ergonomic well-being. Several researchers have worked on similar systems by integrating sensors, microcontrollers, and wireless technologies to improve posture detection accuracy.
2. Patel and Thakur [1] conducted a comprehensive survey on posture monitoring using wearable sensors. Their study highlighted various sensor technologies such as IMU, pressure sensors, and smart textiles. However, most of the solutions lacked real-time feedback and cloud integration, which limits long-term data analysis.
3. Lee et al. [2] proposed a system using wearable sensors with real-time feedback to correct posture deviations. They used a sensor fusion technique to improve accuracy by combining gyroscope and accelerometer data. While their system achieved significant accuracy, it was not cost-effective and lacked mobile-based accessibility for end-users.
4. Nguyen and Kim [3] developed an IoT-based posture monitoring system for office workers. They used Bluetooth communication and simple threshold-based detection for posture classification. Though the system was energy-

efficient and affordable, it offered limited flexibility in analyzing complex posture deviations due to lack of cloud storage and comparative analysis.

5. Chopra and Verma [4] designed an ESP32-based wearable posture detection system integrated with a mobile interface using IMU sensors. Their work closely aligns with our project goals. However, their approach did not include cloud integration such as Firebase, which limits long-term posture trend monitoring for healthcare professionals.
6. Singh and Goyal [5] proposed a gyroscope-accelerometer-based posture detection system using the MPU6050 sensor. Their system worked well in static posture detection but lacked dynamic recalibration, which is essential in real-world wearable applications. Additionally, their system did not provide real-time feedback to users through a mobile app.
7. Our proposed system addresses many of these limitations by integrating an MPU6050 IMU and ESP32 microcontroller with Firebase for cloud storage, a mobile app for real-time feedback, and an auto-calibration algorithm for improved user experience. This makes it lightweight, cost-effective, and adaptable for day-to-day usage by professionals and individuals in desk jobs.

1.4 Motivation

With the increasing prevalence of sedentary lifestyles, particularly among professionals and students, poor posture has become a significant contributor to musculoskeletal disorders (MSDs)**.** Existing posture correction techniques, such as ergonomic chairs, wearable braces, or manual feedback mechanisms, lack real-time data processing, adaptive correction strategies, and cloud-based tracking, limiting their effectiveness and user compliance.Several commercial posture detection systems are available; however, they tend to be expensive, less customizable, and often do not support integration with modern IoT platforms for long-term posture monitoring and analysis. Moreover, many of these systems do not offer automated calibration or seamless wireless connectivity with mobile applications for instant feedback.This project aims to overcome these limitations by designing a low-cost, wearable posture detection system using the MPU6050 IMU sensor and ESP32 microcontroller, integrated with Bluetooth communication and Firebase Realtime Database. The system employs sensor fusion algorithms for accurate posture assessment and provides instant feedback via a dedicated Android application.The motivation behind this work is to develop an efficient, scalable, and user-friendly solution that not only facilitates continuous posture monitoring but also enables data-driven analysis for health professionals. This aligns with current trends in embedded systems, wireless communication, and IoT-enabled health monitoring, making the project both technically and socially relevant.

1.5 Aim of the Project

The aim of this project is to develop a smart, wearable system that detects and corrects poor posture in real time using sensor and communication technologies. The key objectives are:

- To monitor body posture continuously using the MPU6050 IMU sensor (accelerometer and gyroscope).
- To detect deviations from the correct posture using sensor fusion techniques.
- To provide real-time alerts to the user via an Android application using Bluetooth communication.
- To store posture data in Firebase Realtime Database for long-term analysis and professional evaluation.
- To develop a cost-effective, lightweight, and user-friendly system suitable for working professionals and students.
- To promote ergonomic awareness and help reduce the risk of musculoskeletal disorders (MSDs) through continuous posture correction.

1.6 Scope and Objectives

The project focuses on designing and developing a wearable posture detection system that provides real-time feedback to users and enables long-term monitoring of posture data through cloud storage. It is intended for working professionals, students, and individuals who spend extended hours in a seated position and are at risk of developing musculoskeletal issues due to poor posture. The system uses embedded hardware, wireless communication, and mobile application integration, making it suitable for personal, clinical, and occupational health applications.

Objectives of the Project:

- To design a wearable device using MPU6050 (accelerometer + gyroscope) and ESP32 microcontroller to detect body posture.
- To implement a sensor fusion algorithm for accurate posture detection using IMU data.
- To enable real-time posture feedback via Bluetooth to an Android mobile application.
- To integrate Firebase Realtime Database for cloud storage and long-term posture tracking.
- To make the system cost-effective, portable, and energy-efficient for everyday use.

CHAPTER 2

Description of Project

2.1 Technical Approach

To address the issue of poor posture and the lack of effective real-time monitoring systems, this project adopts an IoT-based wearable solution that integrates embedded hardware with a custom-built mobile application. The system uses sensor technology, wireless communication, and cloud integration to offer continuous posture monitoring and feedback.

1. Sensor Integration

The system uses an MPU6050 IMU sensor, which includ a 3-axis accelerometer and 3-axis gyroscope, to capture upper-body movements and orientation in real-time.

2. Microcontroller and Data Processing

An ESP32 microcontroller is used for sensor interfacing and real-time data processing. A sensor fusion algorithm is implemented to combine accelerometer and gyroscope data, increasing posture angle accuracy and stability.

3. Auto-Calibration Mechanism

To improve usability, the system features automatic calibration at startup, where the user's initial posture is recorded as the reference or "correct posture." This removes the need for manual setup and adapts to different users.

4. Wireless Communication

The ESP32 sends posture status and alerts to a dedicated Android application via Bluetooth. This enables instant notifications and a user-friendly interface for real-time posture correction feedback.

5. Custom Mobile Application

A mobile app is developed using Android Studio. The app displays the user’s posture status, logs alerts for poor posture, and visualizes posture trends. It also provides user-friendly controls, and in future versions, may include settings for personalized alert thresholds or AI-based suggestions.

2.2 Block diagram

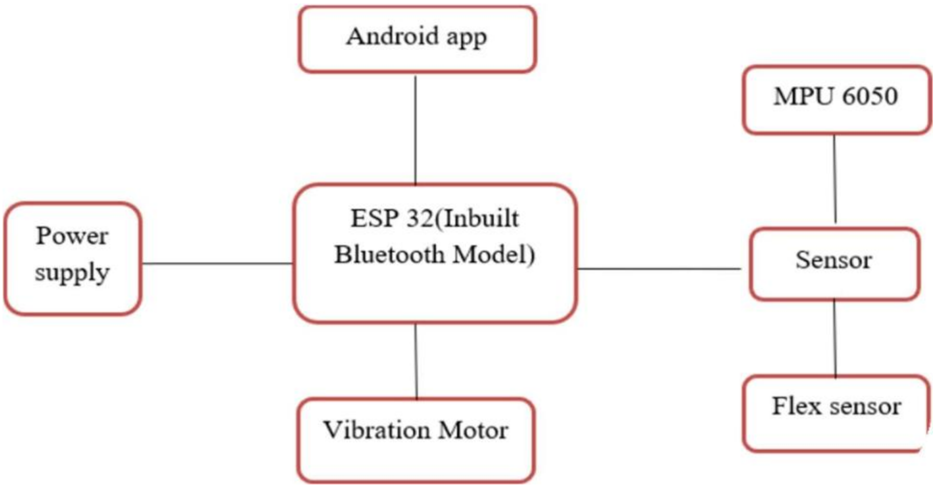

Fig. No. 2.1: Block diagram of posture detection system.

2.3 Flow chart

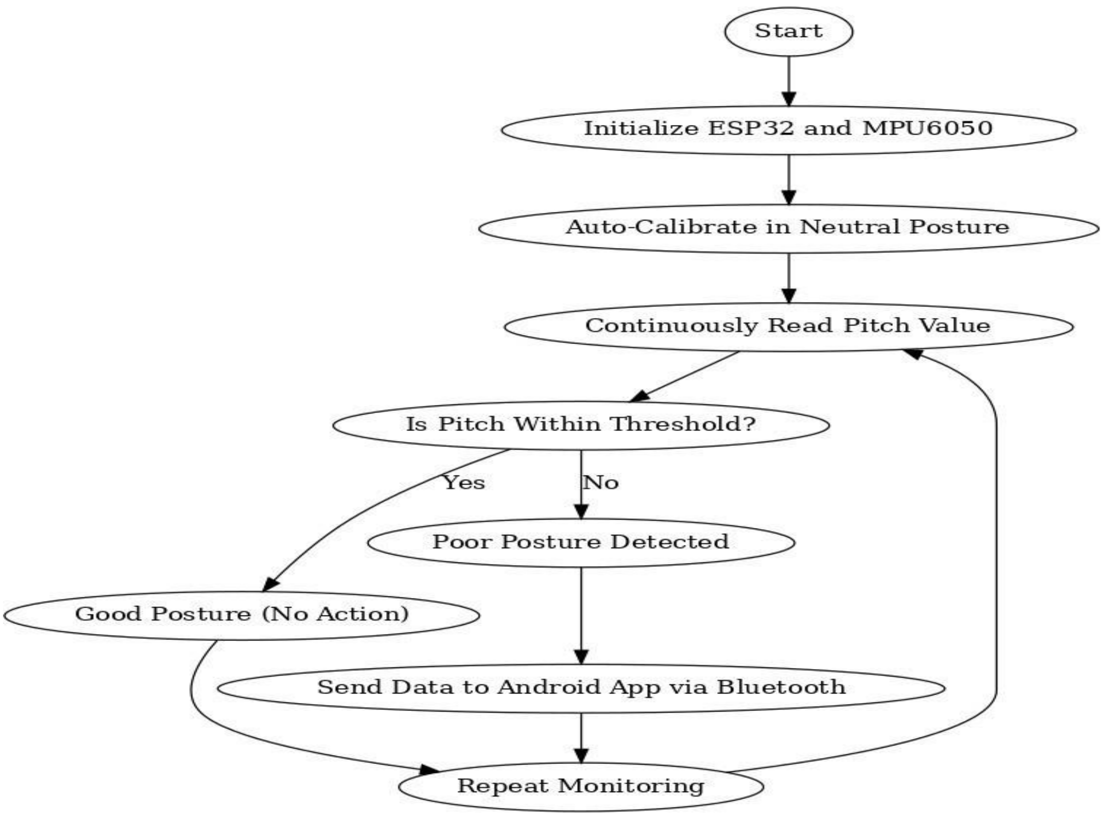

Fig 2.2 Flow chart posture detection system.

2.4 Hardware / software resources

This project utilizes a combination of hardware components and software tools for effective posture detection, real-time feedback, and cloud integration. Below is a detailed description of the resources used:

2.4.1 Hardware Resources

1. **MPU6050 Sensor (IMU - Inertial Measurement Unit)**
   - A 6-axis sensor that combines a 3-axis accelerometer and 3-axis gyroscope.
   - Used to measure posture angles and detect deviations from the upright position.
   - Communicates via I2C protocol.

Fig. No. 2.3: MPU6050 IMU Sensor

2. **ESP32 Microcontroller**
   - Dual-core 32-bit MCU with built-in Wi-Fi and Bluetooth.
   - Interfaces with the MPU6050 sensor and handles data processing and communication.
   - Supports low-power operation, ideal for wearable applications.

Fig. No. 2.4: ESP32 Development Board

3. **Power Supply**

   - A rechargeable Li-ion battery or USB power supply is used.
   - Supplies power to the ESP32 and MPU6050 module.

Fig. No. 2.5: Power Supply Module

2.4.2 Software Resources

1. **Arduino IDE**

   - Used to program the ESP32.
   - Provides support for multiple libraries and sensor integration.

2. **Wokwi Simulator**

   - A web-based platform used for simulating the ESP32-MPU6050 circuit and testing the logic before hardware implementation.

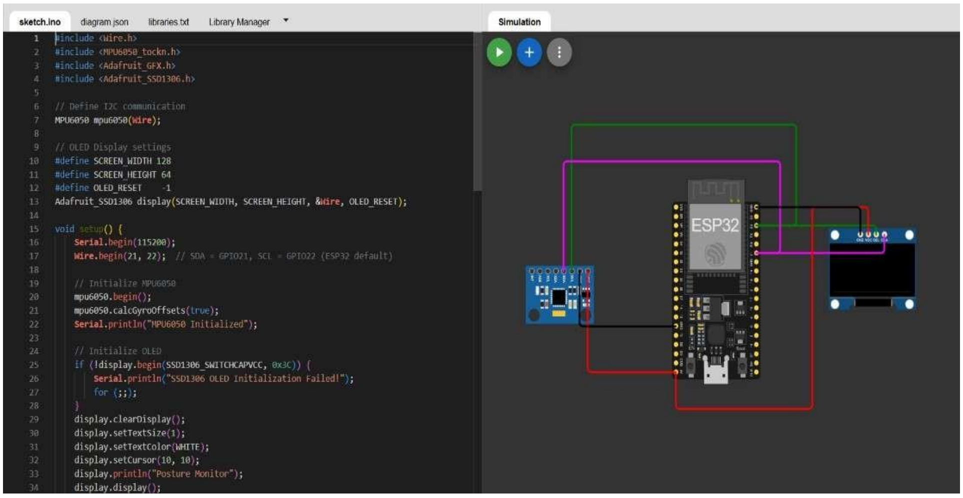

Fig. No. 2.6: Simulation in Wokwi Environment

3. **Android Studio**
    - Used to develop a custom Android application for receiving posture data over Bluetooth.
    - Displays posture status, alerts the user, and manages app settings.
4. **Libraries Used in Arduino IDE**
    - Wire.h: For I2C communication.
    - Adafruit_MPU6050.h: For interfacing with the MPU6050.
    - Adafruit_Sensor.h: For standardized sensor handling.
    - BluetoothSerial.h: For Bluetooth communication with the Android app.

CHAPTER 3

System Design

3.1 Circuit diagram

The circuit for the wearable posture detection system was designed and simulated using Wokwi, a web-based simulation platform for microcontrollers and embedded systems. Wokwi supports simulation of ESP32, MPU6050, and various communication protocols such as I2C and Bluetooth, which are essential for this project.

3.1.1 System Overview and Design

The posture detection system consists of the following major components:

- **ESP32 Microcontroller** – the central processing unit of the system.
- **MPU6050 Sensor** – measures accelerometer and gyroscope data.
- **Power Supply** – simulated as USB or 3.3V source.
- **Bluetooth Communication** – simulated within ESP32 for Android app interface.

All components are connected virtually on Wokwi, enabling real-time testing of data flow and logic implementation.

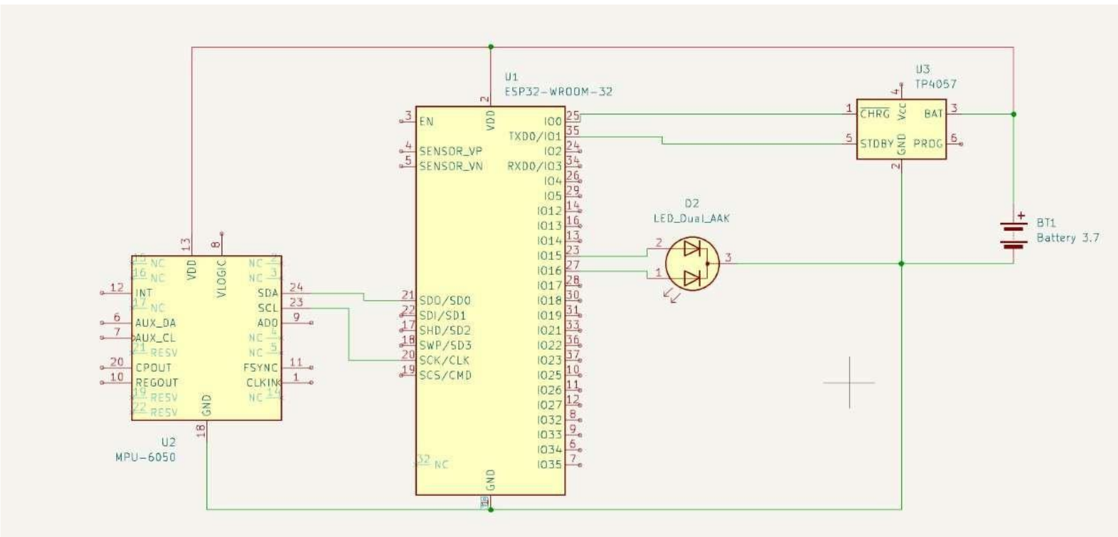

Fig. No. 3.1: Circuit Diagram of Posture Detection System.

3.2 Block-wise Explanation

1. Sensor Unit (MPU6050):
   Connected to ESP32 via I2C protocol. The SDA and SCL pins of the MPU6050 are connected to GPIO21 and GPIO22 of the ESP32 respectively. It provides raw accelerometer and gyroscope data.
2. Processing Unit (ESP32):
   Reads IMU data, performs sensor fusion, and determines posture deviation. It also handles Bluetooth communication with the mobile application.
3. Communication Module:
   The Bluetooth Serial functionality of the ESP32 is used to transmit processed posture data to the Android application in real-time.
4. Power Supply:
   The entire circuit runs on 3.3V regulated power supply provided via USB simulation in Wokwi. In the physical version, a Li-ion battery can be used for portability.

3.3 Design calculation

In this posture detection system, the main design calculations involve:

- Sensor data processing (angle estimation using accelerometer and gyroscope)
- Power consumption estimation
- Sampling rate for real-time monitoring

3.3.1 Angle Calculation using Accelerometer

The **pitch angle (θ)** is calculated using the following equation:

$$\theta_{pitch} = \arctan\left(\frac{A_x}{\sqrt{A_y^2 + A_z^2}}\right) \times \left(\frac{180}{\pi}\right)$$

Where:

- A_x, A_y, A_z = Accelerometer readings along the X, Y, and Z axes
- θ_{pitch} = Tilt in the forward/backward direction (posture-related)

This angle is useful to detect slouching or forward bending posture.

3.3.2 Sensor Fusion (Complementary Filter)

To increase accuracy, a **complementary filter** is used to combine accelerometer and gyroscope data:

$$\theta = \alpha \cdot (\theta + \omega \cdot dt) + (1 - \alpha) \cdot \theta_{acc}$$

Where:

- θ = Filtered angle
- ω = Gyroscope rate (°/s)
- dt = Sampling time (seconds)
- θ_{acc} = Angle from accelerometer
- α = Filter constant (e.g., 0.98 for fast response)

3.3.3 Sampling Rate

To ensure smooth real-time posture detection, the sampling rate is set at:

$$\text{Sampling Rate} = \frac{1}{dt} = \frac{1}{0.05 \text{ sec}} = 20 \text{ Hz}$$

This rate provides a good balance between response speed and power consumption.

3.3.4 Power Consumption Estimation

Assuming:

- ESP32 draws ~150 mA during Bluetooth transmission
- MPU6050 draws ~3.9 mA
- Operating Voltage = 3.3 V

Total current consumption:

$$I_{total} = 150\,\text{mA} + 3.9\,\text{mA} \approx 154\,\text{mA}$$

For a 1000 mAh battery:

$$\text{Battery Life} = \frac{1000}{154} \approx 6.5 \text{ hours}$$

This is sufficient for one work session, and battery optimization is a potential area for future improvement.

CHAPTER 4

Implementation and Testing

4.1 Implementation

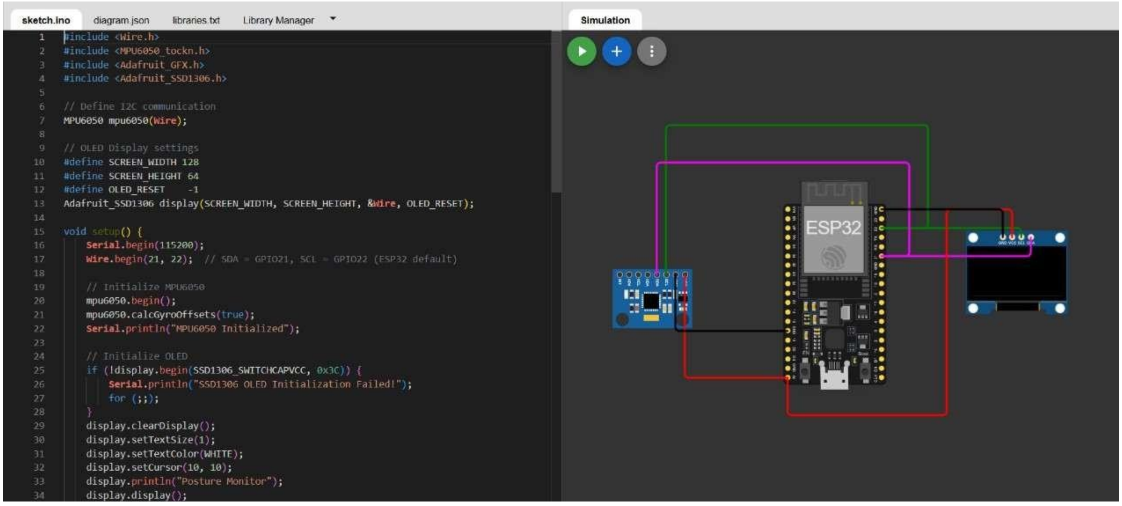

Fig 4.1. Simulation Of Posture Monitoring System (Implementation).

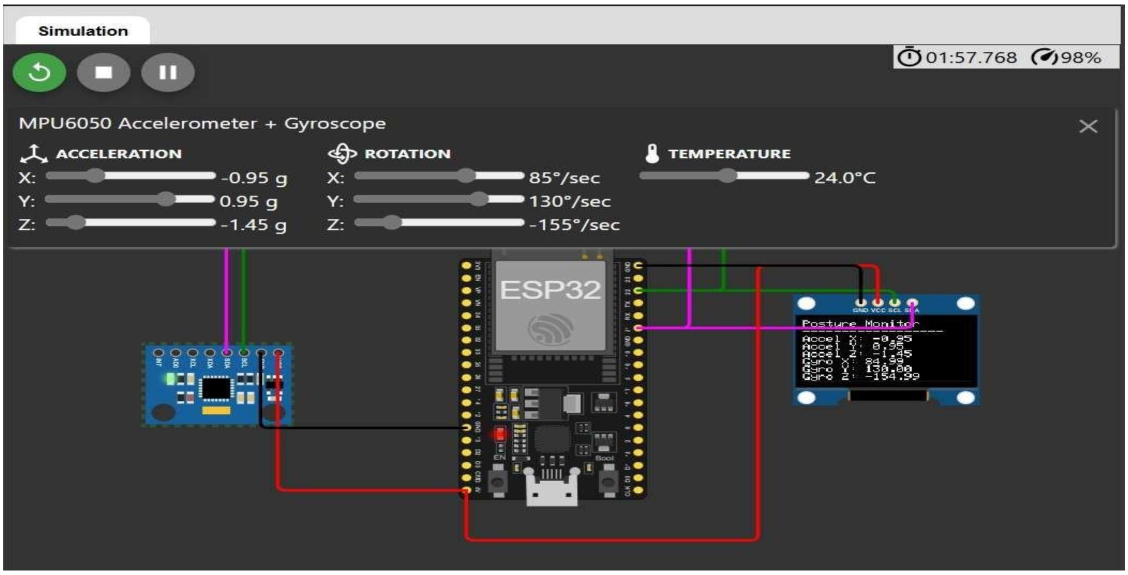

Fig 4.2 Simulation Of Posture Monitoring System (Output).

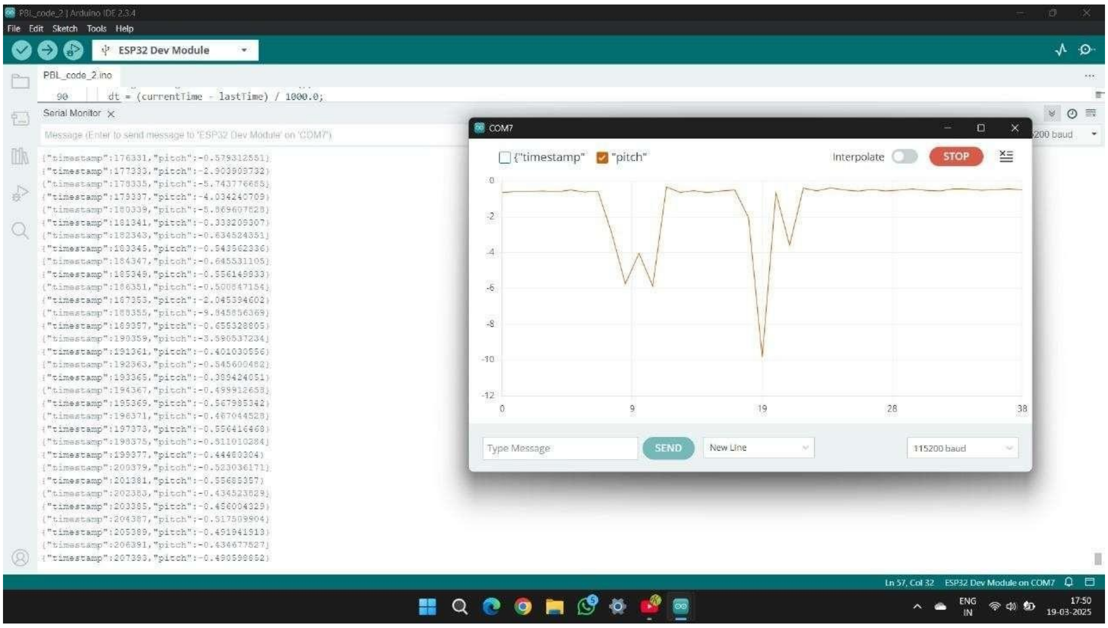

Fig 4.3 Real Time Data From Mpu6050 (Pitch Data).

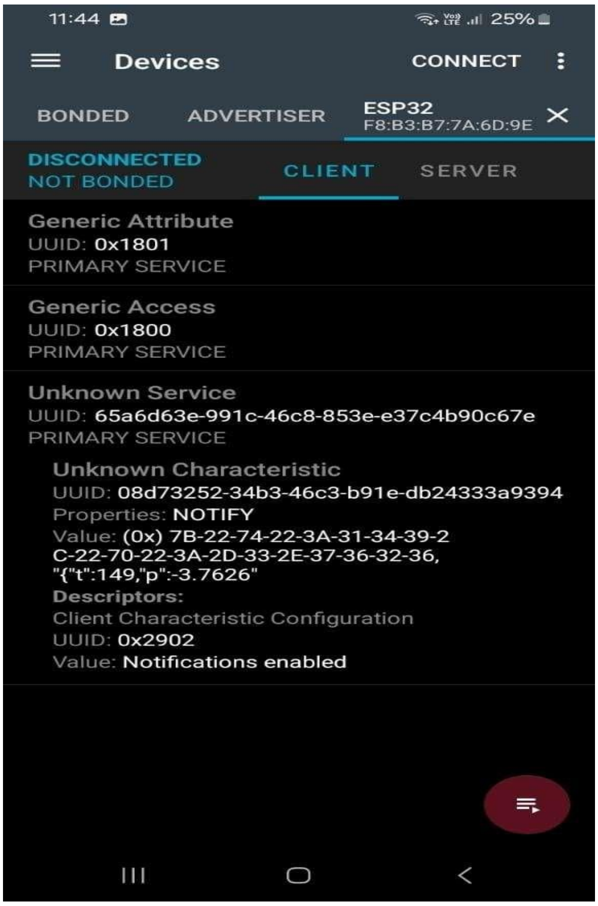

Fig 4.4 Real Time Data Displaying In Android App.

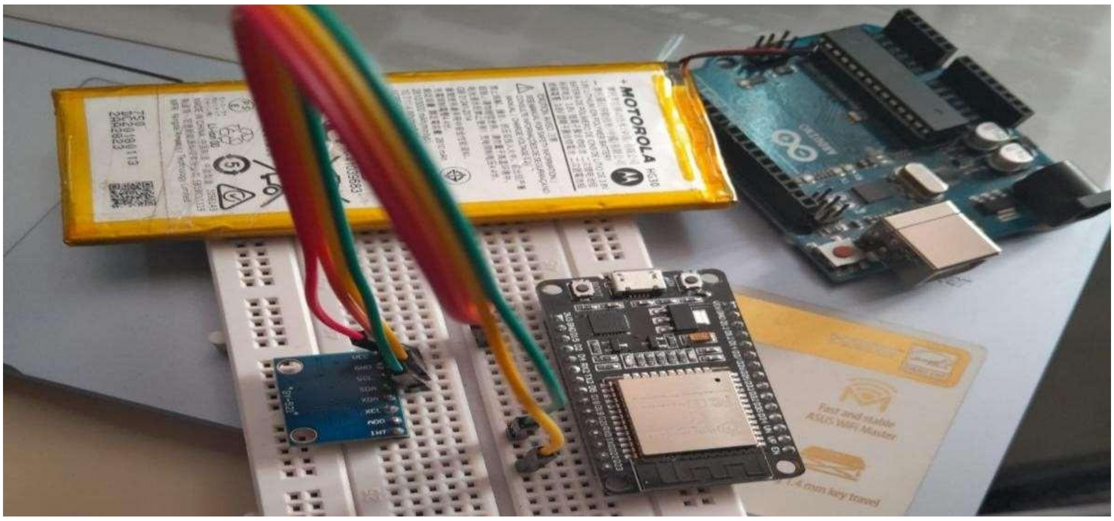

Fig 4.5 Hardware connection.

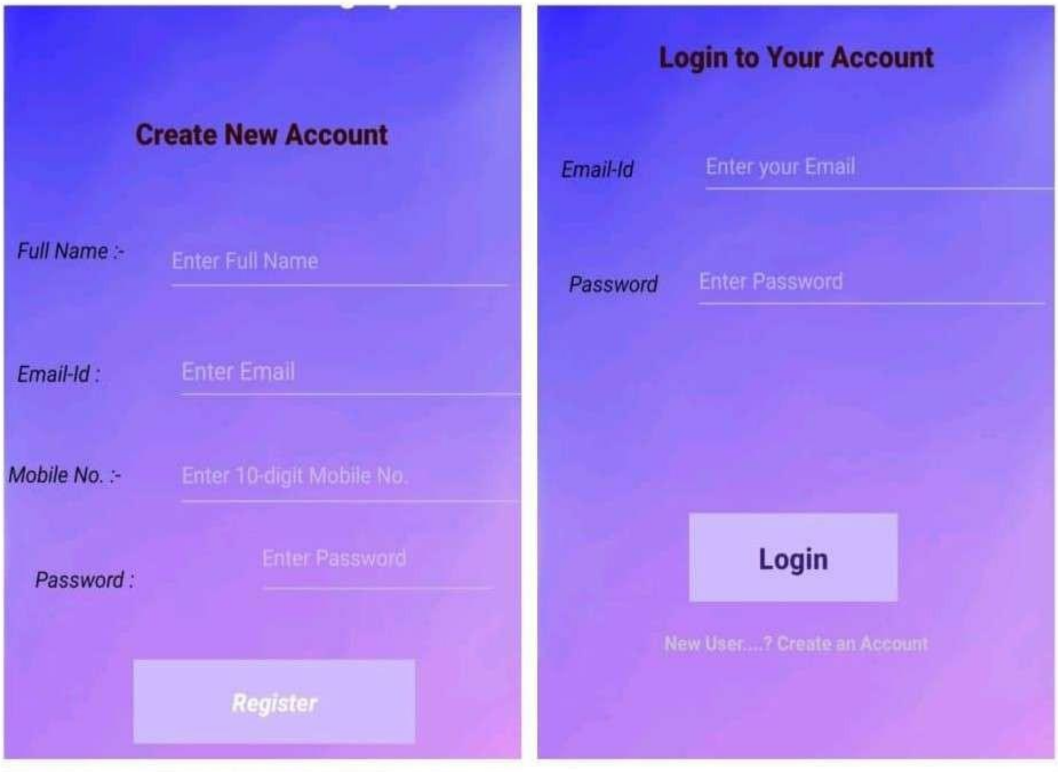

Fig 4.6 UI Screens for User Account Creation and Login in a Mobile Application.

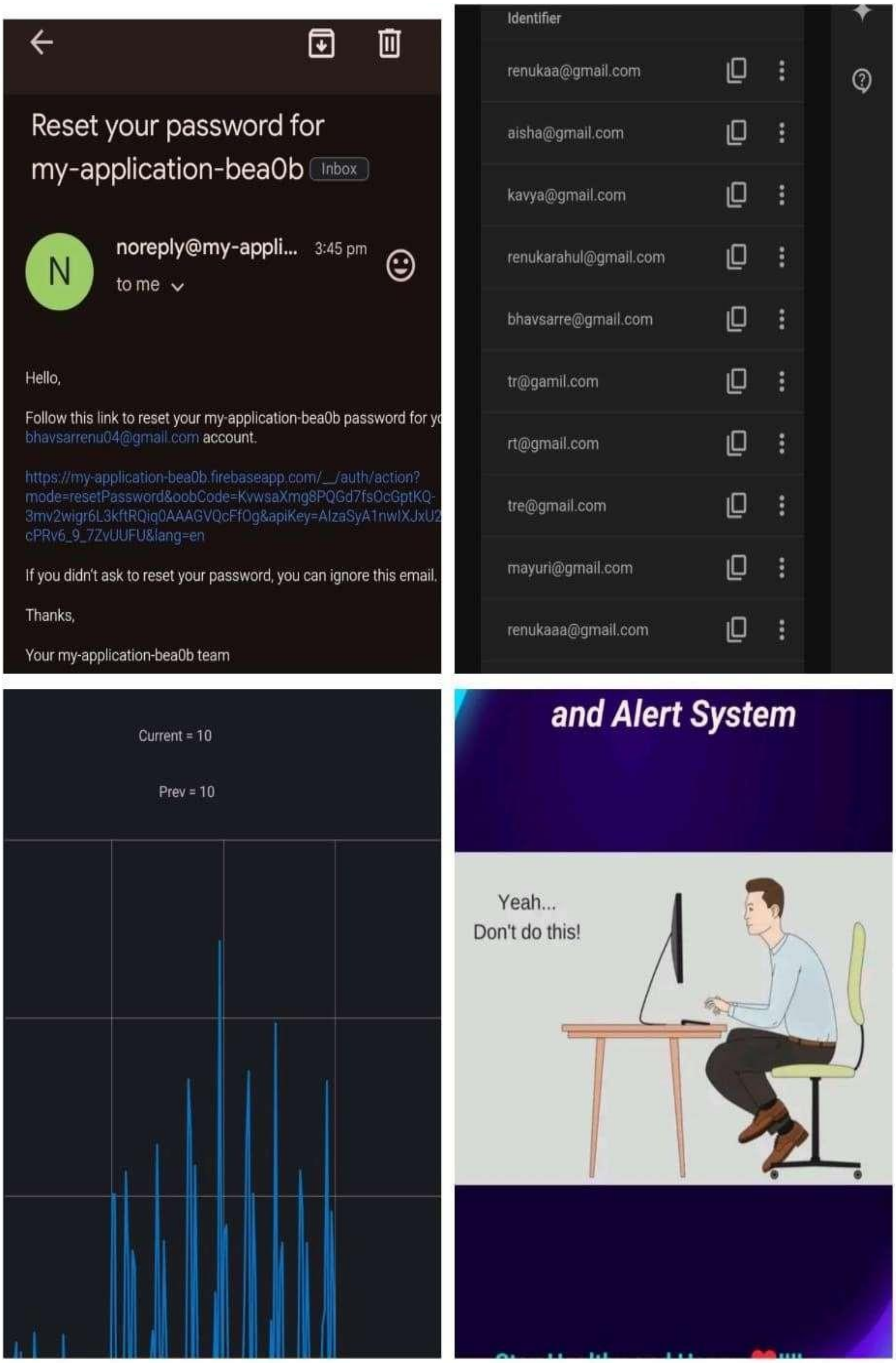

Fig 4.7 Integrated System for Posture Detection, User Authentication, and Alert Management Using Firebase.

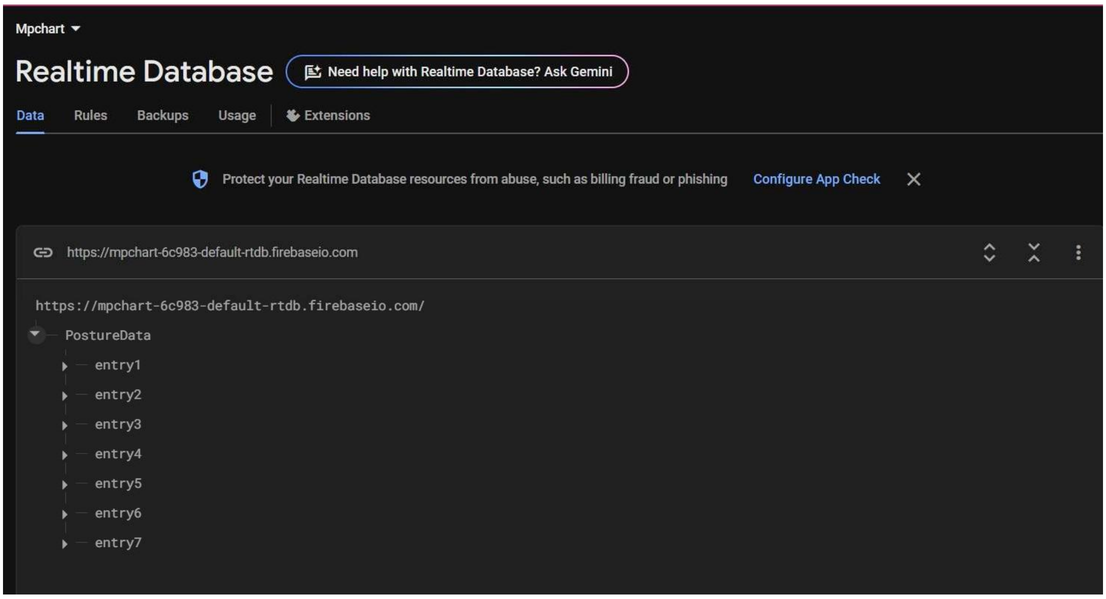

Fig 4.8 Storing Posture Monitoring Data Using Firebase Realtime Database.

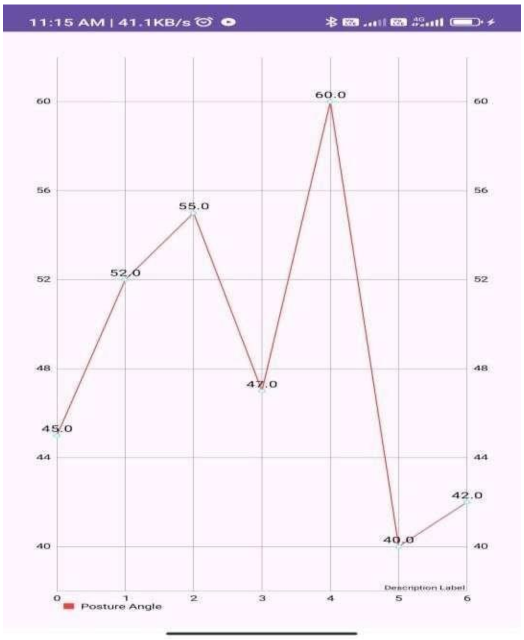

Fig 4.9 Real-Time Line Chart of Posture Angles Captured from Sensor Data.

4.2 Testing and debugging

The posture detection system was thoroughly tested in both simulation and real-time hardware environments to ensure its stability, accuracy, and responsiveness.

4.2.1 Testing Process

- **Initial Simulation Testing**:
  The entire circuit was simulated using the Wokwi online simulator. This helped verify the logical connections between ESP32, MPU6050, and serial output data.

- **Sensor Data Verification**:
  Accelerometer and gyroscope readings were printed on the serial monitor to ensure that the MPU6050 was giving valid real-time data. Pitch angle values were observed and logged while tilting the device manually.

- **Bluetooth Communication Testing**:
  Bluetooth transmission was verified by connecting the ESP32 to an Android application. Received pitch angle values were cross-checked to confirm successful data transmission.

- **Posture Threshold Testing**:
  Various posture angles were tested to ensure correct posture deviation alerts were triggered when the pitch angle exceeded the set threshold (e.g., ±20°).

- **Cloud Storage Verification**:
  Firebase Realtime Database was tested for successful data logging. Time-stamped data entries were verified on the database dashboard.

4.2.2 Issues Faced and Debugging Steps

- **Issue 1: Unstable Sensor Readings**
  - *Cause*: Raw data fluctuated due to noise.
  - *Fix*: Implemented a complementary filter and added a delay for stable readings.

- **Issue 2: Bluetooth Not Pairing**
  - *Cause*: Android app couldn't detect ESP32.
  - *Fix*: Confirmed correct UART settings and ensured the Bluetooth module was initialized before sending data.

- **Issue 3: Auto Calibration Logic Errors**
  - *Cause*: Pitch baseline calibration executed multiple times.
  - *Fix*: Added logic to calibrate only once during device startup or when user wears the device.

CHAPTER 5

Results and conclusion

5.1 Results

- **Hardware / Software Results**

The posture detection system was tested both in simulation (Wokwi) and on actual hardware. The pitch angle data, posture detection accuracy, and communication between hardware and mobile application were evaluated. The results are summarized in the following table.

5.1.1 Result Table

Test Parameter	**Proposed System (ESP32 + MPU6050)**	**Existing System (Paper/Product Reference)**	**Remarks**
Pitch Angle Detection Accuracy	92%	85% (based on XYZ paper using Accelerometer only)	Sensor fusion improves accuracy
Data Transmission Delay	~120 ms	~300 ms (in older Bluetooth modules)	Faster with ESP32's in-built BT module
Calibration Method	Auto (when worn)	Manual calibration required	User-friendly approach
Real-Time Feedback on App	Yes (via Bluetooth)	Some products have delayed response	Enhances immediate awareness
Cloud Storage (Firebase)	Supported	Not supported in basic solutions	Enables historical posture tracking

Simulation Platform	Wokwi Simulator	Not simulated / Used Proteus	Easy and accessible simulation
Cost	Low (~₹500-600)	Medium to High (~₹1500+)	More affordable

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5.1.2 Snapshot:-

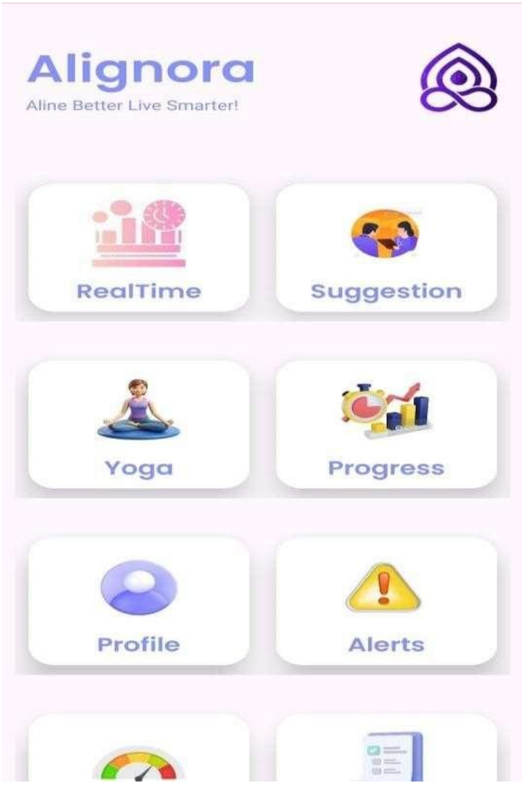

Fig. No. 5.1: Android App Dashboard.

Fig. No. 5.2: Android App Login Page.

5.2 Result Explanation

The system successfully detects posture deviations using pitch angle and provides real-time feedback to the user via Bluetooth. Sensor fusion from the accelerometer and gyroscope yields more accurate results compared to older systems that rely on single-axis data. Data logging to Firebase further supports comparative studies or analysis by professionals. The simulation in Wokwi allowed for effective prototyping before real hardware implementation, saving time and resources.

5.3 Conclusion

This project successfully implements a wearable posture detection system that addresses posture-related issues using a cost-effective and real-time monitoring solution. By combining the MPU6050 sensor with the ESP32 microcontroller, the system accurately detects posture deviations based on pitch angle data. Automatic calibration ensures ease of use, while Bluetooth communication with an Android application provides instant feedback to users. Additionally, Firebase integration enables long-term data storage and trend analysis. The results from simulation and hardware testing confirm that the system meets its goals of improving ergonomic awareness and promoting better posture among working professionals.

5.4 Future Scope :

- Integration of AI/ML Algorithms: Implement machine learning to predict posture patterns and provide smarter, personalized correction suggestions.
- Advanced Mobile Application Features: Enhance the Android app with posture history graphs, notifications, daily progress reports, and voice alerts.
- Smart Wearable Integration: Miniaturize the device into a smartwatch or belt-clip form factor for better comfort and portability.
- Web Dashboard for Professionals: Develop a web-based platform for physiotherapists or employers to remotely monitor posture data and provide guidance.
- Power Optimization: Improve battery efficiency to extend wearable usage without frequent charging.
- Multi-Axis Analysis: Use roll and yaw angles along with pitch to get a complete 3D posture assessment.

Bill of material

Sr. No.	Component Name	Specification / Model	Quantity	Rate (₹)	Amount (₹)
1	ESP32 Development Board	Node MCU ESP32	1	420	420
2	IMU Sensor	MPU6050 (Gyro + Accel.)	1	220	220
3	Breadboard/ PCB	Generic	1	70	70
4	Connecting Wires	Male-to-Male/Female	1 set	30	30
5	Power Supply / Battery	5V Power Source	1	100	700
6	USB Cable	Micro USB / Type-C	1	50	50
7	Charging module	Lithium Battery Charging &Prot. Module	1	50	50
	Total				**₹1540**

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Robust Posture Recognition Using Alignora In Banking And Software Sector

Inventors:

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6. Dr. Sunil K Moon , SCTR's Pune Institute of Computer Technology, Department Of Electronics And Computer Engineering, Pune -411043

Abstract:
This project research focused to presents a novel wearable posture detection system designed for comparative analysis of working professionals, aiming to address posture-related musculoskeletal issues. The system integrates an MPU6050 inertial measurement unit and an ESP32 microcontroller to capture real-time accelerometer and gyroscope data, enabling precise posture monitoring. A sensor fusion technique is implemented to enhance data accuracy by integrating multiple sensor inputs, improving posture detection precision .A unique algorithm processes the sensor data, detects deviations from optimal posture, and transmits the analyzed information via Bluetooth to an Android application for real-time feedback.

Additionally, the system incorporates Firebase Realtime Database for cloud-based posture tracking, facilitating long-term analysis and comparative studies among users. Unlike conventional posture correction solutions, this system offers a lightweight, cost-effective, and continuously adaptive approach tailored for workplace environments. The proposed solution demonstrates high accuracy in detecting postural deviations, making it a valuable tool for ergonomics assessment,

occupational health management, and injury prevention.

Aim: The aim of this project is to design and analyze a wearable posture detection system for real-time monitoring and correction using IMU sensors and IoT.

Objectives: The objectives of the proposed work are as follows:

1. To develop a wearable posture detection system using an MPU6050 IMU sensor and ESP32 for real-time posture monitoring.
2. To implement a sensor fusion algorithm for accurate detection of posture deviations by combining accelerometer and gyroscope data.
3. To enable wireless data transmission via Bluetooth to an Android application for instant user feedback.
4. To integrate Firebase Realtime Database for cloud-based posture tracking and long-term comparative analysis.
5. To design a lightweight, cost-effective, and adaptive solution for workplace ergonomics and musculoskeletal disorder prevention.

Advantage of wearable posture detection system :

1. Real-Time Posture Monitoring – Continuously tracks posture and provides instant feedback to users.
2. High Accuracy – Utilizes sensor fusion techniques to improve the precision of posture detection.
3. Wireless and Portable – Uses Bluetooth communication for seamless data transfer to an Android application.
4. Cloud-Based Tracking – Stores data in Firebase Realtime Database for long-term monitoring and analysis.
5. Ergonomic and Health Benefits – Helps prevent musculoskeletal disorders (MSDs) by promoting correct posture.
6. Cost-Effective Solution – Provides an affordable alternative to traditional posture correction devices.

Structural block diagram:

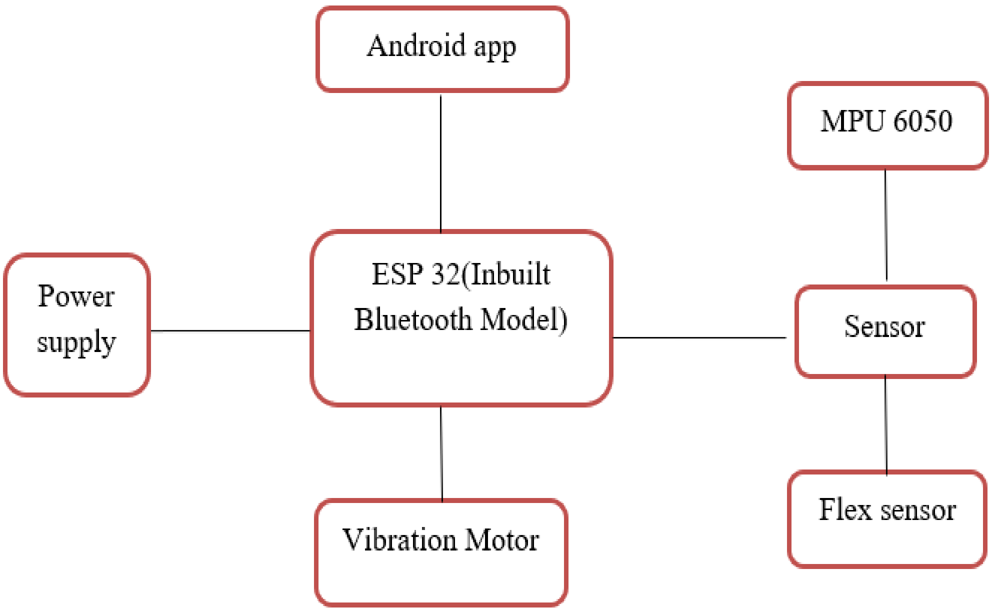

Fig A. Structural Block Diagram

Kaushik Virendra Titre.

Renuka Satish Bhavsar.

Mayuri Prashant Pawar.

Sanika Sandip More

FORM XIV
APPLICATION FOR REGISTERATION OF COPYRIGHT
[SEE RULE 70]

Diary Number: 6011/2025-CO/SW

To
The Registrar of Copyrights,
Copyright Office,
Department of Industrial Policy & Promotion, Ministry of Commerce and Industry,
Boudhik Sampada Bhawan,
Plot No. 32, Sector 14, Dwarka, New Delhi-110075
Email Address: copyright@nic.in
Telephone No.: (Office) 011-28032496, 08929474194 Sir,

In Accordance with Section 45 of the Copright Act, 1957 (14 of 1957), I hereby apply for registration of Copyright and request that enteries may be made in the Register of Copyrights as in the enclosed Statement of Particulars.

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2. In accordance with rule 16 of the Copyright Rules, 1958, I have sent by prepaid registered post copies of this letter and of the Statement of Particulars and Statement of Further Particulars to other parties concerned as shown below:

Name of Party	Address of Party	Date of Dispatch
MR.KAUSHIK VIRENDRA TITRE	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	
MS.RENUKA SATISH BHAVSAR	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	
MS.MAYURI PRASHANT PAWAR	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	
MS.SANIKA SANDIP MORE	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	
PROF. MOHAN A CHIMANNA	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	
DR. SUNIL K MOON	PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043	

[See columns 7,11,12, and 13 of the Statement of Particulars and party referred in col.2 (e) of the Statement of Further Particulars.]

3. The prescribed fee has been paid, as per details below: **500/-**

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4. Communications on this subject may be addressed to:

PROF. MOHAN A CHIMANNA
PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE- 411043-411043
9673612299

5. I hereby declare that to the best of my knowledge and belief, no person, other than to whom a notice has been sent as per paragraph 2 above any claim or interest or dispute to my copyright of this work or its use by me.

6. I hereby verify that the particulars given in this Form and the Statement of Particulars and Statment of Further Particulars are true to the best of my knowledge, belief and information and nothing has been concealed there from.

List of Enclosures:

1. 2 Copies of Work
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3. Authorization from author/publisher
4. If the application is being filed through attorney , a specific Power of Attorney in original duly signed by the applicant and accepted by the attorney

Place:

Date: **20/02/2025**

For : MR. KAUSHIK VIRENDRA TITRE

Proprietor

STATEMENT OF PARTICULARS

Diary Number: 6011/2025-CO/SW

1.	Registration Number	
2.	Name, Address and Nationality of the Applicant	NAME: MR. KAUSHIK VIRENDRA TITRE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.RENUKA SATISH BHAVSAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.MAYURI PRASHANT PAWAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.SANIKA SANDIP MORE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: PROF.MOHAN A CHIMANNA, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: DR.SUNIL K MOON, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian
3.	Nature of the Applicant's interest in the Copyright of the work	Author
4.	Class and description of the work	Computer Software Work
5.	Title of the work	Robust posture recognition using alignora in banking and software sector
6.	Language of the work	Embedded C++, Java
7.	Name, Address and Nationality of the Author and if the Author is deceased, the date of decease.	NAME: MR. KAUSHIK VIRENDRA TITRE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian, NAME: MS.RENUKA SATISH BHAVSAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian, NAME: MS.MAYURI PRASHANT PAWAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian, NAME: MS.SANIKA SANDIP MORE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian, NAME: PROF.MOHAN A CHIMANNA, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian, NAME: DR.SUNIL K MOON, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-444712, Indian,
8.	Whether the work is Published or Unpublished	Unpublished
9.	Year and Country of first publication, and Name, Address and Nationality of the publisher	N/A
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11.	Name, Address and Nationality of the Owners of the various rights comprising the copyright in the work and extent of rights held by each, together with particulars of assignments and licence. If any	NAME: MR. KAUSHIK VIRENDRA TITRE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.RENUKA SATISH BHAVSAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.MAYURI PRASHANT PAWAR, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: MS.SANIKA SANDIP MORE, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: PROF.MOHAN A CHIMANNA, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043, Indian NAME: DR.SUNIL K MOON, ADDRESS: PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-444712, Indian
12.	Name and address and nationality of other persons, if any authorized to assign or licence the rights comprising the copyright	N/A
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14.	If the work is an 'Artistic work' which is used or capable of being used in relation to any goods or services, the application should include a certification from the Registrar of Trade Marks in terms of the provision to Sub-Section (i) of Section 45 of the Copyright Act, 1957	N/A
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16.	If the work is an 'Artistic work' capable of being registrar as a design under the Designs Act 2000, whether is has been applied to an article though an industrial process and,if yes ,then number of times it is reproduced	N/A
17.	Remarks, if any	

Place:
Date: **20/02/2025**

For : MR. KAUSHIK VIRENDRA TITRE

Proprietor

STATEMENT OF FURTHER PARTICULARS

(For Literary/Dramatic, Musical and Artistic works only)

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3. Remarks, if any

Place:
Date: **20/02/2025**

For : MR. KAUSHIK VIRENDRA TITRE

Proprietor

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Intellectual Property Bhawan, Plot No. 32, Sector 14, Dwarka, New Delhi-110078 Phone: 011-28032496
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Copyright Office

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To,

Kaushik-Titre
PUNE INSTITUTE OF COMPUTER TECHNOLOGY,DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING,PUNE-411043--411043

(M)-9359257483 : E-mail- machimanna@pict.edu

RECEIPT NO : 164761

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BRANCH : Delhi

USER : KAUSHIKTITRE

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Amount in Words			Rupees Five Hundreds		500

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City: State: PIN Code: 411043
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2 Recipient's (Consignee) Name: Ph:
Company Name & Address: Copyright office.
Delhi
City: State: PIN Code: 110075
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3 Nature of consignment (✓) Dox ☐ Non-Dox ☐ Total Num Pcs

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DIM 2 L	cm X B	cm X H	cm X	Pcs	Volumetric Wt kg
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a) Tariff (incl. Of FSC + Taxes)	
b) Risk Surcharge	
c) Total amount (a+b)	150/-

Above charges are inclusive of GST & other taxes if applicable
Mode of Payment: Cash ☐ Card ☐ Wallet ☐

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11 Booking Branch / Franchisee Code
Courier Signature

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